Filter Engine Instruction Set
Quick Reference

ON Semiconductor®

M-20789-004
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General Instruction Format:
`INS` *[optional codes]* `dst, src,` *<other arguments>*

Operand Definitions

ID	Operand	Description	Variable Range
sc	SC**x**	State code for conditional processing	**x** = 0-15
dst-sc	SC**x**	Destination state code (as the output of CMP instruction)	**x** = 0-15
sh	LEFT**x**, RIGHT**y**	Shift value	**x** = 1-3, **y** = 1-4
dst-reg src-reg-1 src-reg-2	E**w** S**x** IN**y** OUT**z**	Data register	**w** = 0-15 **x** = 0-27 **y** = 0-11 **z** = 0-7
c	C**x**	Coefficient set	**x** = 0-31
sv	SV**x**	State variable set	**x** = 0-31
qf	RND, ESS	Quantizer format	RND = Rounding ESS = Error spectral shaping
cond	EQ, NE, GT, LT, GTE, LTE, ZERO, NOT_ZERO	Condition codes	EQ = (src1==src2) NE = (src1!=src2) GT = (src1 > src2) LT = (src1<src2) GTE = (src1≥src2) LTE = (src1<=src2) ZERO = (src1==0) NOT_ZERO = (src1!=0)
int	NI, CI, SI, TI	Interrupt control settings	NI = No interrupt CI = Clear interrupt SI = Set interrupt TI = Transition interrupt
atc rtc	Integer constant **x**	Attack or release time constants expressed as bit shift by **x**+2	**x** = 0-15
ad	NM, SM	Absolute value disable (to optionally use the level detector as a smoothing filter)	SM = Smoothing filter enable NM = Normal level detector

Instruction Set

Mnemonic	Description	Instruction Format	Examples
ABS	Absolute value	ABS *[sc, sh]* dst-reg, src-reg-1	abs e10, e5 abs [sc5, right2] s4, e3
ADD	Addition	ADD *[sc, sh]* dst-reg, src-reg-1, src-reg-2	add e4, e2, e3 add [sc1, left1] s2, e2, in2
BIQ	Biquad filter	BIQ *[sc]* dst-reg, src-reg-1, c, sv, qf	biq e7, e9, c3, sv7, rnd
CMP	Arithmetic comparison	CMP *[sc]* dst-sc, src-reg-1, src-reg-2 or 0, cond, int	cmp sc0, e0, e1, gt, si cmp [sc3] sc3, e0, e1, gte, ci cmp sc2, e4, 0, zero, ni
FIR	FIR filter	FIR *[sc]* dst-reg, src-reg-1, c, sv	fir e0, s2, c0, sv0 fir s15, e0, c2, sv2
LDET	Level detector	LDET *[sc]* dst-reg, src-reg-1, atc, rtc, ad	ldet s15, s0, 4, 6, nm ldet e1, s0, 14, 14, sm
LMLA	Multiply src-reg-2 with inverse log2 of src-reg-1 and add to dst-reg	LMLA *[sc, sh]* dst-reg, src-reg-1, src-reg-2	lmla s4, s1, s17
LMLS	Multiply src-reg-2 with inverse log2 of src-reg-1 and subtract from dst-reg	LMLS *[sc, sh]* dst-reg, src-reg-1, src-reg-2	lmls s4, s1, s17
LMUL	Multiply src-reg-2 with inverse log2 of src-reg-1 and put in dst-reg	LMUL *[sc, sh]* dst-reg, src-reg-1, src-reg-2	lmul s4, s1, s17
LOG	Base-2 logarithm	LOG *[sc]* dst-reg, src-reg-1	log [sc5] s1, e3
MLA	Multiply src-reg-1 and src-reg-2 and add to dst-reg	MLA *[sc, sh]* dst-reg, src-reg-1, src-reg-2	mla [sc15, left1] e2, s1, e2
MLS	Multiply src-reg-1 and src-reg-2 and subtract from dst-reg	MLS *[sc, sh]* dst-reg, src-reg-1, src-reg-2	mls [sc6, left1] e0, in2, e0
MOV	Move data from src-reg-1 to dst-reg	MOV *[sc, sh]* dst-reg, src-reg-1	mov s3, s9 mov [left2] s3, e5
MUL	Multiplication	MUL *[sc, sh]* dst-reg, src-reg-1, src-reg-2	mul e3, in1, e2
NOP	No operation	NOP	nop
SUB	Subtraction	SUB *[sc, sh]* dst-reg, src-reg-1, src-reg-2	sub [sc4] e12, e4, e8